

*One of the most powerful tools in data management
and analysis.*

Microsoft EXCEL



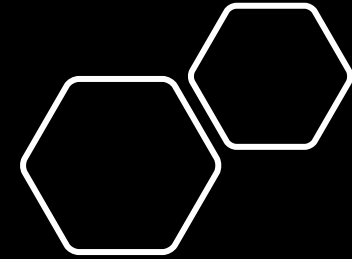
Manoj Pandey

An aerial photograph of a winding road on a mountain. The road is paved and curves through lush green forests and rocky terrain. In the background, there are more mountains and a river. The overall scene is misty and scenic.

Master Excel Formulas|

Master Excel formulas are advanced functions that combine multiple operations to perform complex calculations and data analysis. They often use nested functions, array formulas, and lookup techniques to manipulate large datasets efficiently. These formulas enable users to automate tasks, create dynamic reports, and extract valuable insights from spreadsheets. Mastering them requires practice but greatly enhances productivity and analytical capabilities.

Unlocking the Power of Excel Formulas ={{(🔒)}}



AGENDA



Financial Formula

- ☰ PMT



Text Formula

- ☰ Upper
- ☰ Lower
- ☰ Proper
- ☰ Right
- ☰ Left
- ☰ Mid
- ☰ Concatenate
- ☰ Len
- ☰ Text & Substitute
- ☰ Trim



Look & Reference

- ☰ Vlookup
- ☰ Xlookup
- ☰ Index
- ☰ Match
- ☰ Unique & Transpose



Logical Formula

- ☰ True
- ☰ False
- ☰ If
- ☰ Iferror
- ☰ Ifs
- ☰ And, Or, Not



Statistical Formula

- ☰ Average
- ☰ Min
- ☰ Max
- ☰ Count
- ☰ CountA
- ☰ Count Blank
- ☰ Countif
- ☰ Countifs
- ☰ Median & Mod



Math Formula

- ☰ Subtotal
- ☰ Sum & Sumproduct
- ☰ Sumif & Power



Date & Time

- ☰ Day, Month, Year
- ☰ Edate & Eomonth
- ☰ Datedif & Today
- ☰ Networkdays
- ☰ Now, Time, Hour
- ☰ Minute & Second

Financial Formula

“Calculating Loan Payments with the PMT Formula”



Calculations:



EMI (Equated Monthly Installment): $=\text{PMT}(\text{B3}/\text{B5}, \text{B5}*\text{B4}, -\text{B2})$

Calculates monthly payments based on annual interest rate (B3), number of payments per year (B5), loan term in years (B4), and loan amount (B2).



Total Repayment Amount: $=\text{B7}*\text{B5}*\text{B4}$

Total amount repaid over the loan term



Total Interest Amount: $=\text{B8}-\text{B2}$

Total interest paid over the loan term.



Apply:

Input:

	A	B	C	D	E
1	Home Loan				
2	Loan Amt	₹1,550,000.00			
3	Interest rate	3%			
4	Years of Loan	20			
5	Repayment Mode	12			

Formula:

8	Total Repayment Amt	$=\text{B7}*\text{B5}*\text{B4}$
7	EMI	$=\text{PMT}(\text{B3}/\text{B5}, \text{B5}*\text{B4}, -\text{B2})$
9	Total Interest Amt	$=\text{B8}-\text{B2}$

Output:

7	EMI	₹8,596.26
8	Total Repayment Amt	₹2,063,103.06
9	Total Interest Amt	₹513,103.06

Text Formula Syntax

Upper
=UPPER(Text)

01

02

Lower
=LOWER(Text)

Proper
=PROPER(Text)

03

04

RIGHT
=RIGHT(text,
[num_chars])

Left
=LEFT(text,
[num_chars])

05

06

Mid
=MID(text, start_num,
num_chars)

Concatenate
=CONCATENATE(text1,
[text2], ...)

07

08

Len
=LEN(Text)

Text
=TEXT(value,
format_text)

09

10

Substitute
=SUBSTITUTE(text,
old_text, new_text,
[instance_num])

Trim
=TRIM(Text)

11



Text Formulas

“Transforming and Managing Data with Text Functions”

Formulas:

 **UPPER** formula converts all lowercase to uppercase.

 **Lower** formula converts all uppercase to lowercase.

 **Proper** capitalizes the first letter of each word in a text string.

 Extracts a specified number of characters from the **Right** side of a text string.

 Extracts a specified number of characters from the **Left** side of a text string.

 Extracts a specified number of characters from the **Middle** of a text string, starting at a specified position.

Apply:

1	Input	Formula	Output
2	manoj pandey	=UPPER(A2)	MANOJ PANDEY

1	Input	Formula	Output
2	MANOJ PANDEY	=LOWER(A2)	manoj pandey

	Input	Formula	Output
	MANOJ PANDEY	=PROPER(A2)	Manoj Pandey
	manoj pandey	=PROPER(A3)	Manoj Pandey

	Input	Formula	Output
	OpenAI	=RIGHT(A2,2)	AI

	Input	Formula	Output
	Nvidia	=LEFT(A2,3)	Nvi

	Input	Formula	Output
	Codebasics	=MID(A2,5,3)	bas

Text Formulas

“Transforming and Managing Data with Text Functions”

Formulas:

 **CONCATENATE** Combines multiple text strings into one string.



Input	Formula	Output
OpenAI	<code>=CONCATENATE(A2," ",A3)</code>	OpenAI Microsoft
Microsoft		

 **LEN** Returns the number of characters in a text string and spaces also.



Input	Formula	Output
CONTRA COGS	<code>=LEN(A2)</code>	11

 **TEXT** Converts a value to text in a specific number format.



Input	Formula	Output
1456.456782	<code>=TEXT(A2,"0.00")</code>	1456.46
28-07-24	<code>=TEXT(A3,"ddd-dd-mmm-yyyy")</code>	Sun-28-Jul-2024

 **TRIM** Removes extra spaces from a text string, leaving only single spaces between words and no space characters at the start or end of the text.



Input	Formula	Output
Finance	<code>=TRIM(A2)</code>	Finance

 **SUBSTITUTE** used to replace specific text in a text string with different text. It's particularly useful when you want to replace text without using find and replace manually.



Input	Formula	Output
Tiger	<code>=SUBSTITUTE(A2,"Tiger","Lion")</code>	Lion

Apply:



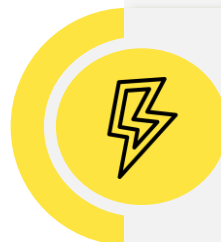
Lookup Formula Syntax



=VLOOKUP(lookup_value,
table_array, col_index_num,
[range_lookup])



=XLOOKUP(lookup_value,
lookup_array, return_array,
[if_not_found], [match_mode],
[search_mode])



=INDEX(return_range,
MATCH(lookup_value,
lookup_range, 0))



=UNIQUE(array, [by_col],
[exactly_once])



{=TRANSPOSE(array)}
Ctrl+Shift+Enter

Look & Reference Formulas

Efficient Data Retrieval and Management Techniques

Formulas:

 **Vlookup** Searches for a value in the leftmost column of a table and returns a value in the same row from a column you specify.



Apply:

	A	B	C	D	E	F
1	Model	Price		Input	Samsung Galaxy M30	
2	Apple iPhone 11	23958		Formula	=VLOOKUP(E1,\$A\$1:\$B\$8,2,0)	
3	Apple iPhone X	21832		Output	20762	
4	Samsung Galaxy M30	20762				
5	Samsung Galaxy Note 8	18948				
6	Realme X	19924				
7	Realme XT	17954				
8	OnePlus 7 Pro	23951				

 **Xlookup** A newer function that improves upon VLOOKUP and HLOOKUP by providing a more flexible and powerful way to look up data in a range.



	A	B	C	D	E	F	G
1	Model	Price		Input	19924		
2	Apple iPhone 11	23958		Formula	=XLOOKUP(E1,B1:B8,A1:A8)		
3	Apple iPhone X	21832		Output	Realme X		
4	Samsung Galaxy M30	20762					
5	Samsung Galaxy Note 8	18948					
6	Realme X	19924					
7	Realme XT	17954					
8	OnePlus 7 Pro	23951					

Look & Reference Formulas

Efficient Data Retrieval and Management Techniques

Formulas:

 **Index** Returns a value or the reference to a value from within a table or range.

Match Returns a value or the reference to a value from within a table or range.

 **Transpose** rotate data in a range of cells from horizontal to vertical or vice versa.

 **Unique** Extracts a list of unique values from a range or array.

Apply:

	A	B	C	D	E	F	G	H	I
1	SalesMan	Date	product	Amt		Input	Oppo		
2	Rohit	01-Sep	Samsung	15,389		Formula	=INDEX(A1:A7,MATCH(G1,C1:C7,0))		
3	Mitali	02-Sep	Nokia	17,111		Output	Bhashkar		
4	Bhashkar	03-Sep	Oppo	14,802					
5	Anumeet	04-Sep	Samsung	10,757					
6	Raunak	05-Sep	Nokia	15,592					
7	Suhail	06-Sep	Lava	15,785					



F	G	H
Input	Formula	Output
=TRANSPOSE(F1:H1)		
Formula		
Output		



After apply this formula hit enter, then press F2 then hit together ctrl+shift+enter. Formula look like

fx  {=TRANSPOSE(F1:H1)}

	A	B	C
1	Input	Formula	Output
2	Samsung	=UNIQUE(A2:A7)	
3	Nokia	Nokia	
4	Oppo	Oppo	
5	Samsung	Lava	
6	Nokia		
7	Lava		



After apply this formula hit enter, then press F2 then hit together ctrl+shift+enter. Formula look like

fx  {=UNIQUE(A2:A7)}

TRUE equals 1, and
FALSE equals 0

=IF(condition,
value_if_true,
value_if_false)

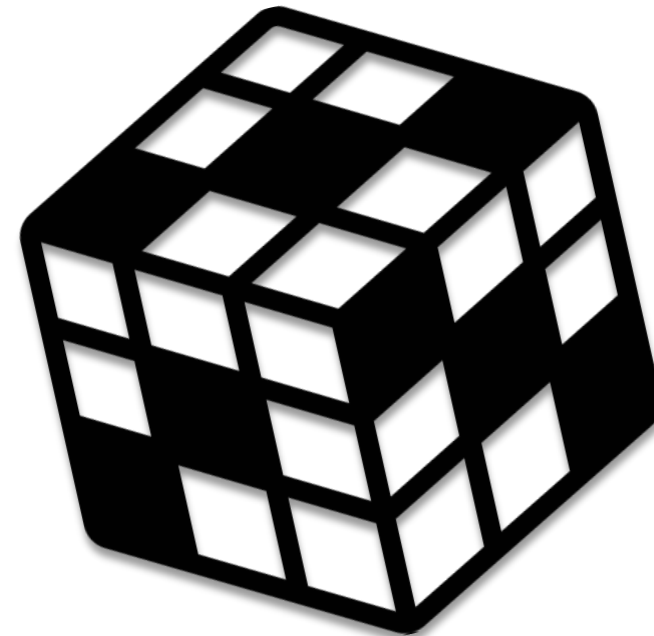
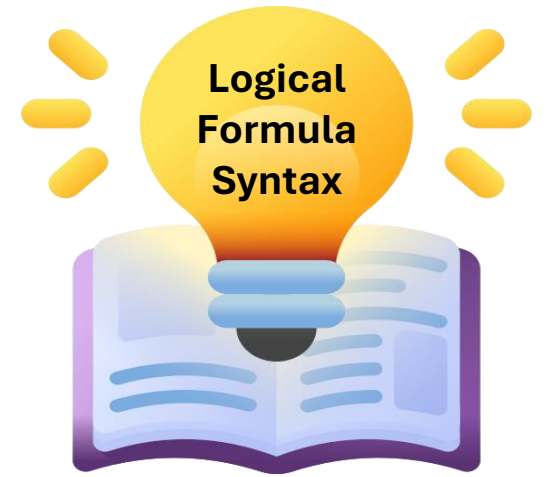
=IFERROR(value,
value_if_error)

=IFS(condition1, value1,
[condition2, value2], ...)

=AND(condition1,
[condition2], ...)

=OR(condition1,
[condition2], ...)

=NOT(condition)



Logical Formulas

Streamlining Decision-Making and Conditional Analysis



Formulas:



True & False These are not formulas but are built-in logical values that represent true and false conditions. TRUE equals 1, and FALSE equals 0.



IF function checks whether a condition is met and returns one value if true and another value if false.



IFERROR function returns a specified value if a formula evaluates to an error; otherwise, it returns the result of the formula.



Apply:

	A	B	C
1	Data		Formula
2	123abc	123abc	=A2=B2
3	098ZXY	098xyz	FALSE

	A	B	C
1	Emp	Salary	Increment 10% if salary < 10000
2	John	9551	10506.1
3	Silva	12897	=IF(B3<10000,B3+B3*10%,B3)
4	Reo	13827	13827
5	Leo	11586	11586

A	B	C	D	E	F
Model	Price	Formula			
Apple iPhone 11	23958	=IFERROR(VLOOKUP(A2,A2:B6,2,0),"not found")			
Apple iPhone X	21832	21832			
	19924	not found			
	17954	#N/A			
OnePlus 7 Pro	23951	23951			

Logical Formulas

Streamlining Decision-Making and Conditional Analysis

Formulas:

 **IFS** function checks multiple conditions and returns a value that corresponds to the first TRUE condition.



Student Name	Marks	Grade
John	100	=IFS(B18<30,"D",B18<=50,"C",B18<=80,"B",B18<=100,"A")
Silva	70	B

 **AND** function returns TRUE if all the conditions are true; otherwise, it returns FALSE.



A	B	C	D	E	F	G	H	I
Sales Person	LED Lamps	Clock	Total Sales	Commision				
John	18702	84886	167197	=IF(AND(B2>=50000,C2>=50000),D2*2%,"Work Hard")				
Silva	59764	58347	192329	3846.58				

 **OR** function returns TRUE if any of the conditions are true; otherwise, it returns FALSE.

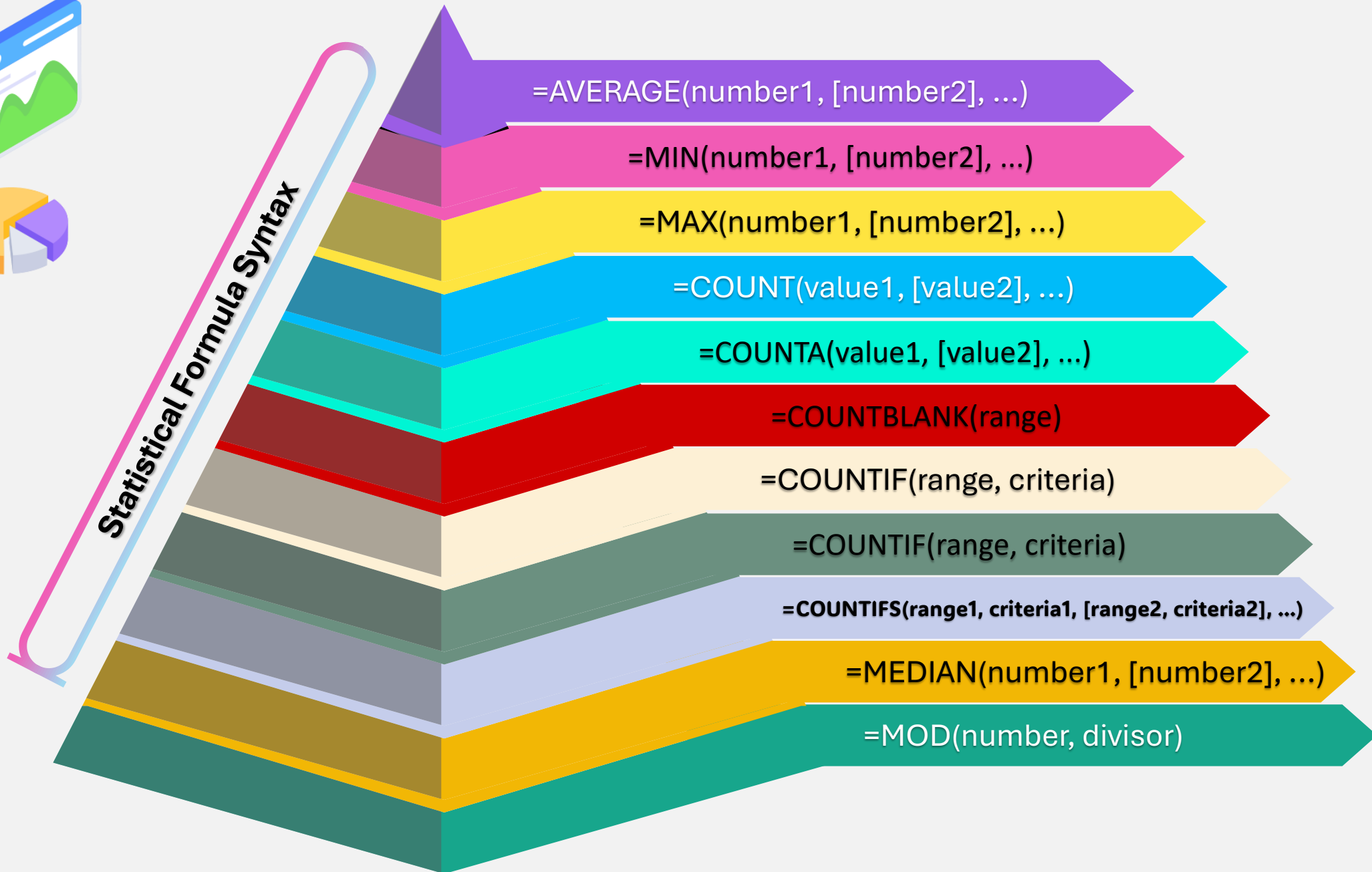


Sales Person	LED Lamps	Clock	Total Sales	Commision				
John	18702	84886	167197	=IF(OR(B2>=50000,C2>=50000),D2*2%,"Work Hard")				
Silva	25870	38619	91026	Work Hard				

 **NOT** function reverses the value of its argument. If the argument is TRUE, NOT returns FALSE; if the argument is FALSE, NOT returns TRUE.



A	B	C	D	E	F
Data	Formula				
Red	=IF(NOT(A2="Red"),"Values are different","Values are the same")				
Blue	Values are different				



Statistical Formulas

Advanced Techniques for Data Analysis and Interpretation

Formulas:

 **Average** Calculates the average of a range of numbers.



 **Min** Returns the smallest number in a range of numbers.



 **Max** Returns the largest number in a range of numbers.



 **Count** Returns the largest number in a range of numbers.



Apply:

1	Name	Sub 1	Sub 2	Formula	
2	John	75	57	=AVERAGE(B2,C2)	
3	silva	45	85		65

Name	Sub 1	Sub 2	Formula	Min No
John	75	57	=MIN(B2:B5,C2:C5)	45
silva	45	85		
Reo	79	45		
Leo	85	63		

Name	Sub 1	Sub 2	Formula	Max No
John	75	57	=MAX(B2:B5,C2:C5)	85
silva	45	85		
Reo	79	45		
Leo	85	63		

Name	Sub 1	Sub 2	Formula	No Count
John	75	57	=COUNT(B2:B5,C2:C5)	8
silva	45	85		
Reo	79	45		
Leo	85	63		

Statistical Formulas

Advanced Techniques for Data Analysis and Interpretation

Formulas:

 **CountA** Counts the number of non-empty cells in a range.



Name	Sub 1	Sub 2	Formula	No Count
John	75	57	=COUNTA(B2:B5,C2:C5)	5
silva		85		
Reo	79			
Leo		63		

 **Countblank** Counts the number of empty cells in a range.



Name	Sub 1	Formula	No Count
John	75	=COUNTBLANK(B2:B5)	2
silva			
Reo	79		
Leo			

 **Countif** count the number of occurrences cells in a range that meet a single condition.



Name	Formula	No Count
John	=COUNTIF(A2:A7,A2)	2
John	1	1
Leo	1	1
Reo	2	2
Reo	1	1
silva	1	1

Statistical Formulas

Advanced Techniques for Data Analysis and Interpretation



Formulas:



Countifs count the number of occurrences cells in a range that meet a multiple criteria.



Apply:

Region	Builder		Region	Builder	Formula	Count
South	John		East	John	=COUNTIFS(A2:A7,D2,B2:B7,E2)	1
East	John					
North	Leo					
South	Reo					
West	Reo					
Central	silva					



Median Returns the median (middle value) of a given set of numbers.



	A	B	C
1	Number	Formula	Output
2	5	=MEDIAN(A2:A11)	19
3	8		
4	12		
5	15		
6	18	Middle Value '=(18+20)/2	
7	20		
8	25		
9	30		
10	35		
11	40		



Mod Returns the remainder after a number is divided by a divisor.



	A	B	C
1	Number	Formula	Output
2	5	=MOD(A3,A2)	3
3	8		

Math Formula Syntax

=SUBTOTAL(func
tion_num, ref1)

=SUMPRODUCT
(array1,[array2],
...)

=POWER(num
ber, power)
=SQRT(number)

=RANDBETWEEN(b
ottom, top)
=ROUND(number,
num_digits)

=ROUNDUP(nu
mber,num_dig
its)

=SUM(number1,
[number2], ...)

=SUMIF(range,
criteria,
[sum_range])

=SUMIFS(sum_range,
criteria_range1,criteri
a1, [criteria_range2,
criteria2], ...)

=ROUNDDOWN
(number,num_d
igits)

=PRODUCT(nu
mber1,
[number2], ...)

$f(x)$

5⁴3²4

Math Formulas

Precision and Power in Mathematical Calculations



Formulas:



Subtotal Calculates the subtotal of a range of cells using a specified aggregate function.



Apply:



Sum Adds all the numbers in a range of cells.



Sumproduct Returns the sum of the products of corresponding ranges or arrays.



=SUBTOTAL(		A	B
N SUBTOTAL(function num, ref1, ...)		1	=SUBTOTAL(9,A3:A12)
5	(...) 1 - AVERAGE	2	Number
8	(...) 2 - COUNT	3	5
12	(...) 3 - COUNTA	4	8
15	(...) 4 - MAX	5	12
18	(...) 5 - MIN	6	15
20	(...) 6 - PRODUCT	7	18
25	(...) 7 - STDEV.S	8	20
30	(...) 8 - STDEV.P	9	25
35	(...) 9 - SUM	10	30
40	(...) 10 - VAR.S	11	35
	(...) 11 - VAR.P	12	40
	(...) 101 - AVERAGE		

Number										
5	8	12	15	18	20	25	30	35	40	=SUM(A2:J2)

	A	B	C	D	E
1	ITEM	QTY	RATE	AMOUNT	
2	1-A	11	95	1045	
3	2-A	13	230	2990	
4	3-A	17	173	2941	
5	4-A	22	150	3300	
6	5-A	21	183	3843	
7	6-A	20	205	4100	
8	GRAND TOTAL			18219	
9	Sumproduct	18219		=SUMPRODUCT(B2:B7,C2:C7)	

Math Formulas

Precision and Power in Mathematical Calculations



Formulas:



Apply:



Sumif Adds the cells specified by a given condition.



	A	B	C	D	E	F	G
1	Region	sales					
2	north	6881		north	=SUMIF(A2:A11,D2,B2:B11)		
3	west	4361		west	22484		
4	east	6946		east	22640		
5	south	4347		south	42349		
6	west	18123					
7	south	13338					
8	north	4185					
9	south	6196					
10	south	18468					
11	east	15694					



Power Returns the result of a number raised to a power.



Number	Formula
8	=POWER(K2,2)
3	9



Sumifs Adds the cells specified by multiple conditions or criteria.

Region	State	sales			
north	punjab	6881	punjab	=SUMIFS(C2:C11,A2:A11,"north",B2:B11,E1)	
west	kolkata	4361	bihar	22640	
east	bihar	6946	goa	36153	
south	goa	4347			
west	chennai	18123			
south	goa	13338			
north	punjab	4185			
south	delhi	6196			
south	goa	18468			
east	bihar	15694			

Math Formulas

Precision and Power in Mathematical Calculations

Formulas:

 **Sqrt** Returns the square root of a number.



 **Randbetween** Returns a random integer between two specified numbers.



 **Round** Rounds a number to a specified number of digits.



 **Rounddown** Rounds a number down, toward zero.



 **Roundup** Rounds a number down, toward zero.



 **Product** Multiplies all the numbers given as arguments.



Apply:

Number	Formula	Output
30	=SQRT(K2)	5

Number	Formula	Output
8	=RANDBETWEEN(1,10)	8

Number	Formula	Output
3.14159	=ROUND(3.14159, 2)	3.14

Number	Formula	Output
3.14159	=ROUNDDOWN(K2,2)	3.14

Number	Formula	Output
3.14159	=ROUNDUP(K2,2)	3.15

Qty	Amt	Formula	Output
5	75	=PRODUCT(K2,L2)	375
6	45	270	270
2	105	210	210
9	20	180	180

Data & Time Formula Syntax

=DAY(date)

=MONTH(date)

=YEAR(date)

=EDATE(start_date, months)

=EOMONTH(start_date, months)

=TODAY()

=DATEDIF(start_date, end_date, unit)

=NETWORKDAYS(start_date, end_date, [holidays])

=NOW()



Date & Time Formulas

Effective Techniques for Date and Time Calculations

Formulas:

 **Day** Returns the day of the month (1-31) from a date value.



 **Month** Returns the month (1-12) from a date value.



 **Year** Returns the year from a date value.



 **Edate** Returns the date that is the specified number of months before or after the start date.



 **Eomonth** Returns the last day of the month that is the specified number of months before or after the start date.



 **Today** Return the current date based on system's date and time.



Apply:

Date	Formula	Output
08-Aug-24	=DAY(A2)	8

Date	Formula	Output
08-Aug-24	=MONTH(A2)	8

Date	Formula	Output
08-Aug-24	=YEAR(A2)	2024

Date	Formula	Output
08-Aug-24	=EDATE(A2,1)	08-09-24

Date	Formula	Output
08-Aug-24	=EDATE(A2,-1)	08-07-24

Date	Formula	Output
08-Aug-24	=EOMONTH(A2,2)	31-10-24

Formula	Output
=TODAY()	07-08-24

Date & Time Formulas

Effective Techniques for Date and Time Calculations

Formulas:

 **Datedif** Calculates the number of days, months, or years between two dates.



 **Networkdays** Returns the number of whole working days between two dates, excluding weekends and specified holidays.



 **Now** Returns the current date and time.



Apply:

DOB	Formula	Output
12-Feb-90	=DATEDIF(A2,TODAY(),"Y")	34 Year
12-Feb-90	=DATEDIF(A2,TODAY(),"YM")	5 Month
12-Feb-90	=DATEDIF(A2,TODAY(),"MD")	27 Day

Joining date	Resign Date		
12-Feb-20	15-Aug-22		
Holidays		Formula	Output
04-02-21		=NETWORKDAYS(A2,B2,A5:A9)	654
25-02-21			
11-03-24			
12-03-24			
13-03-24			

Formula	Output
=NOW()	08-08-24 21:14

Date & Time Formulas

Effective Techniques for Date and Time Calculations

Formulas:



Apply:

 **Time** Returns a decimal number representing a time.



Formula	Output
<code>=TIME(2,15,58)</code>	2:15 AM

 **Hour** Returns the hour (0-23) from a time value.



Time	Formula	Output
12:40:00	<code>=HOUR(\$A\$2)</code>	12:00 AM

 **Minute** Returns the minute (0-59) from a time value.



Time	Formula	Output
13:05:28	<code>=MINUTE(\$A\$2)</code>	5

 **Second** Returns the second (0-59) from a time value.



Time	Formula	Output
13:05:28	<code>=SECOND(\$A\$2)</code>	28



Top Five Key Points:

1



**Data Analysis
and Visualization**

2



**Advanced Formulas
and Functions**

3



**Data Cleaning and
Preparation**

4



**Automation with
Macros and VBA**

5



**Collaboration and
Sharing**

THANK YOU

Your time and attention are greatly appreciated. I hope you found this presentation valuable.

I'd love to hear your thoughts!

- **Feedback:** What did you find most useful or interesting?
- **Comments:** Do you have any questions or insights to share?
- **Improvement:** How can I enhance future presentations? Your suggestions are welcome!

Your input is important to me, and it will help in refining and improving future content. Thank you for sharing your feedback!